BBC NEWS ARTICLES TEXT MINING PROJECT

CLASSIFICATION AND CLUSTERING OF NEWS ARTICLES

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Project Overview

- ❖ Text mining pipeline for **classification** and **clustering**
- ❖ BBC News articles as case study
- ❖ End-to-end workflow:
    - Data loading
    - Preprocessing
    - Text representation
    - Supervised & unsupervised learning

Project Objectives

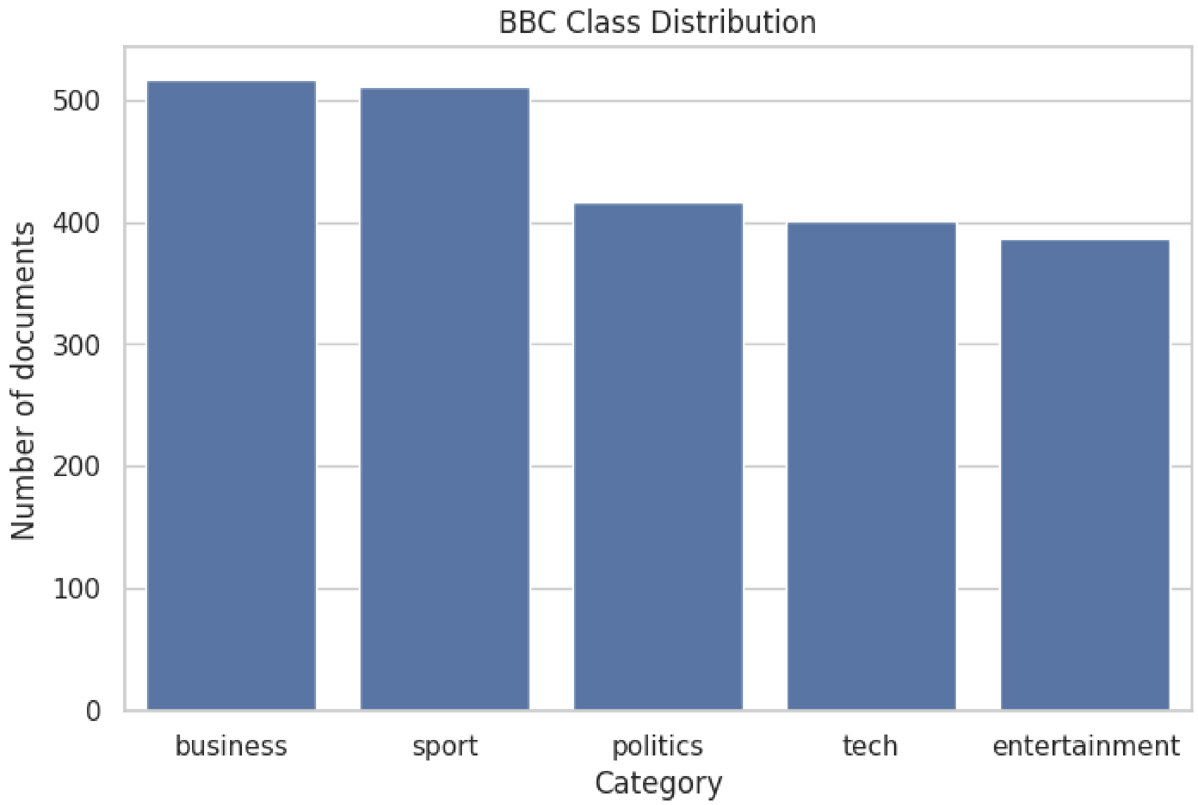
- ❖ Preprocess raw textual data to reduce noise

- ❖ Compare different **text representations**

- ❖ Evaluate:
    - classification performance
    - clustering structure and quality

- ❖ Analyze impact of representations on results

BBC News Dataset



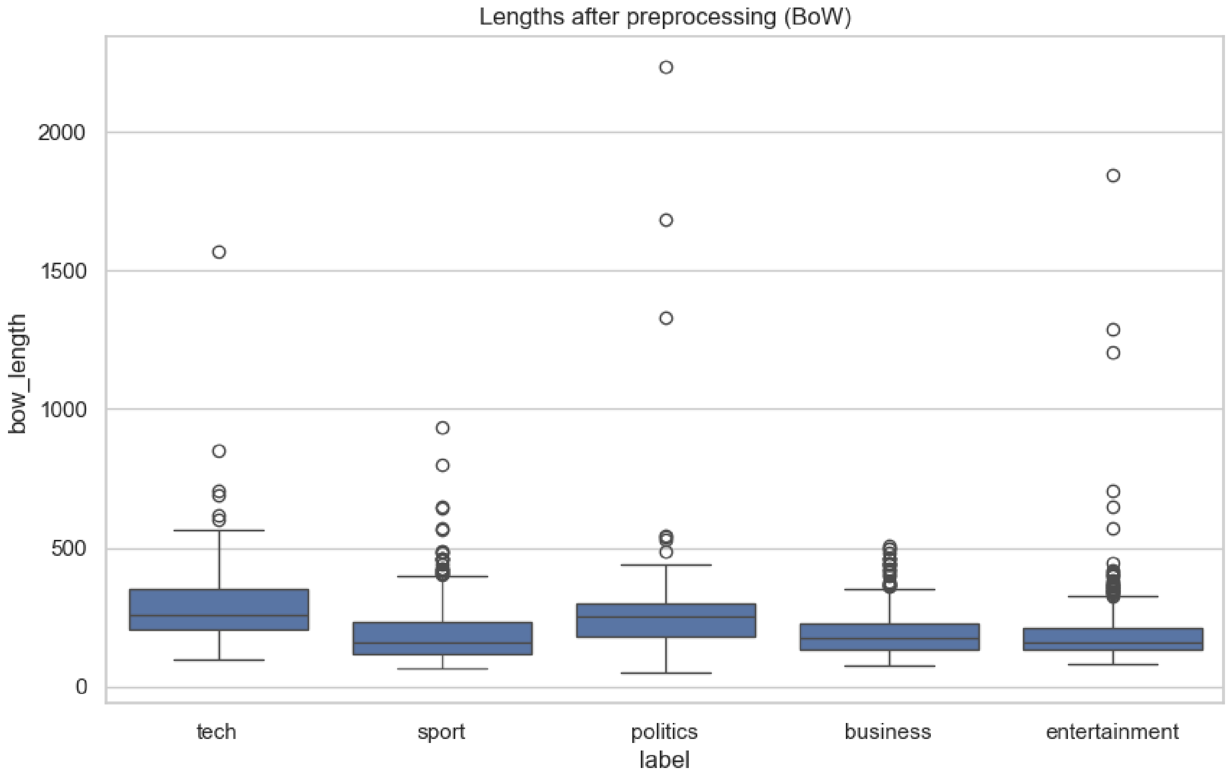
- ❖ **2232 NEWS ARTICLES**
- ❖ ***5 categories:***
  - Business
  - Entertainment
  - Politics
  - Sport
  - Tech
- ❖ Balanced distribution across classes
- ❖ Suitable for both:
  - Classification
  - clustering

Text Preprocessing Strategy

- ❖ Raw text contains noise and inconsistencies
- ❖ Preprocessing depends on the **text representation**
- ❖ Two different pipelines:
  - **Strong preprocessing** → BoW / TF-IDF
  - **Light preprocessing** → BERT embeddings

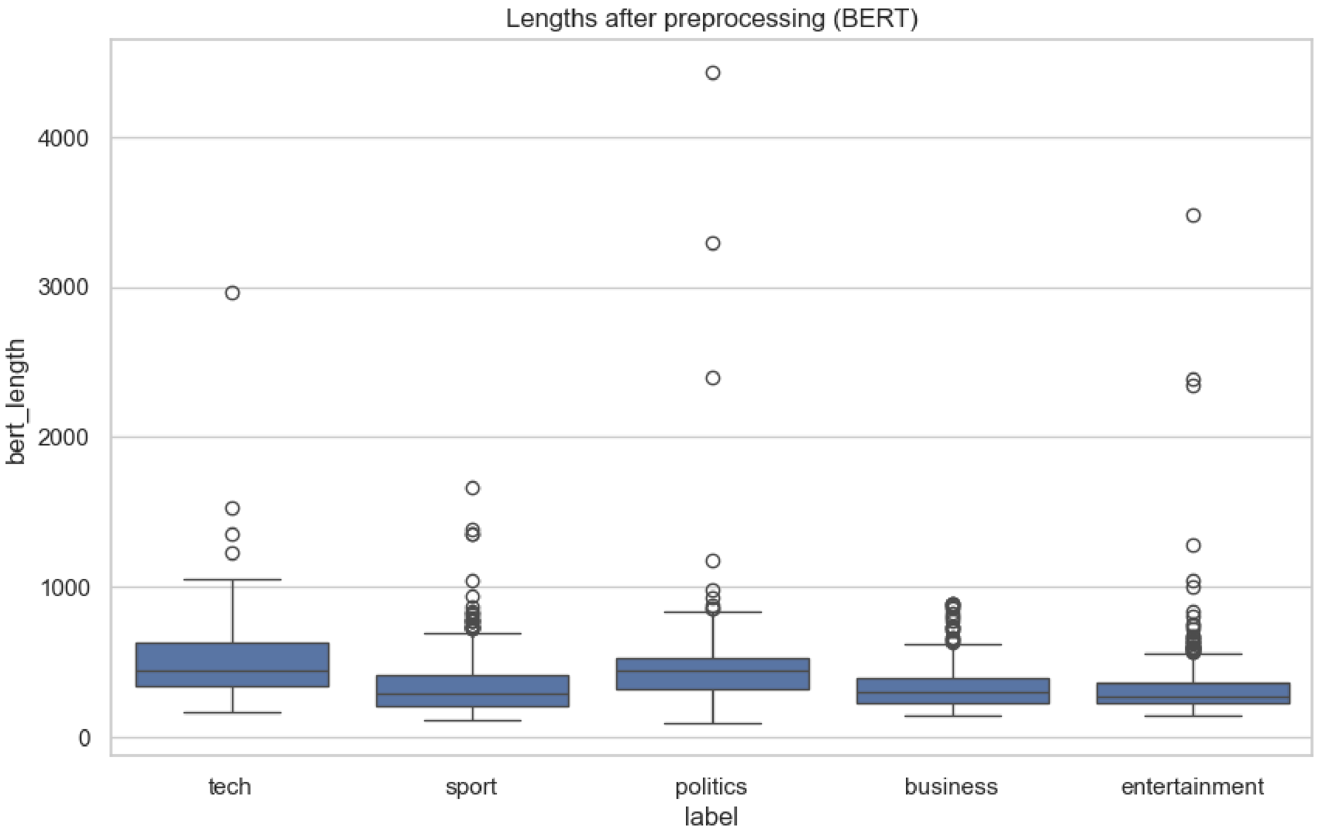
Strong Preprocessing for BoW / TF-IDF

- ❖ Designed for BoW and TF-IDF models
- ❖ Goals:
  - Reduce noise
  - Reduce vocabulary size
  - Improve discriminative power
- ❖ Steps:
  - Lowercasing
  - Punctuation & number removal
  - Stopword removal
  - Lemmatization (spaCy)



Light Preprocessing for Contextual Embeddings (BERT)

- ❖ BERT requires near-original text
- ❖ Aggressive cleaning can harm semantic information
- ❖ Only minimal preprocessing:
  - Newline removal
  - Extra whitespace cleaning
- ❖ No:
  - Lowercasing
  - Lemmatization
  - Stopword removal



Text Representation: TF-IDF

- ❖ Sparse vector representation based on term frequency
- ❖ Strong preprocessing applied to reduce noise
- ❖ Two TF-IDF versions:
    - **Classification TF-IDF** (train / test split)
    - **Full-corpus TF-IDF** (clustering)
- ❖ Interpretable and effective baseline for text-based models

Item	Description
Input	Clean_text_bow → TF - IDF
Output	(2232, 6985)
Saved files	X_all_tfidf.joblib Tfidf_vectorizer_all.joblib (with artefacts included)

Contextual Embeddings with Sentence Transformers

- Dense semantic representations based on a **pretrained transformer model**
- Model used: **all-MiniLM-L6-v2**
- Captures **contextual and semantic meaning** beyond word frequency
- Suitable for:
    - semantic classification
    - document clustering
- Requires **light preprocessing** to preserve original text structure

Item	Description
Input	Clean_text_bert → SentenceTransformer
Output	(2232, 384)
Saved files	embeddings_all.npy

Text Classification — Setup & Inputs

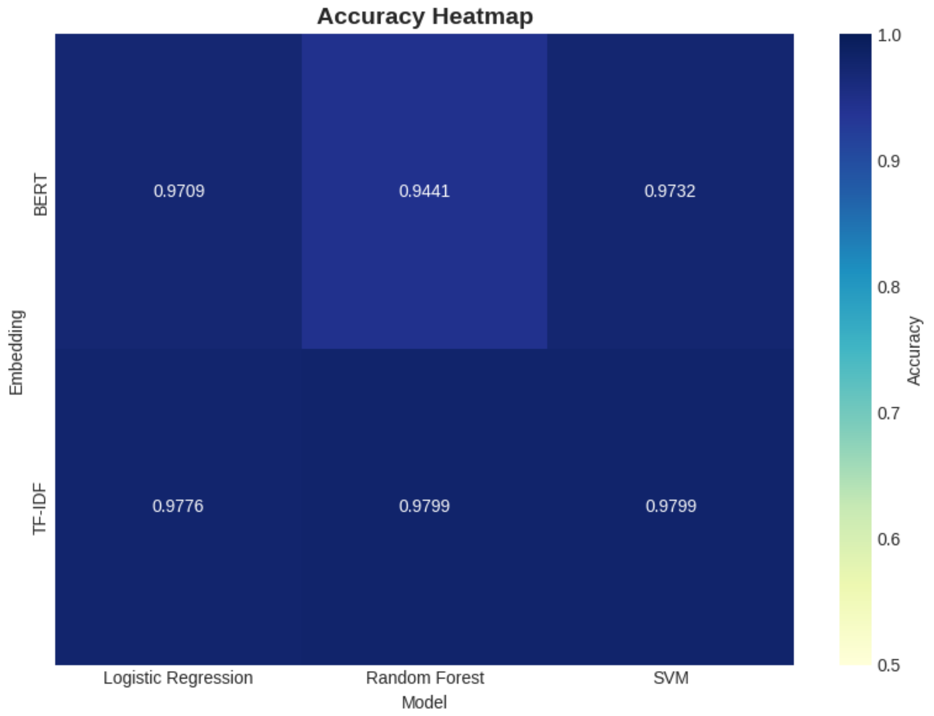
- **Dataset:** BBC News (5 classes), **N = 2232**

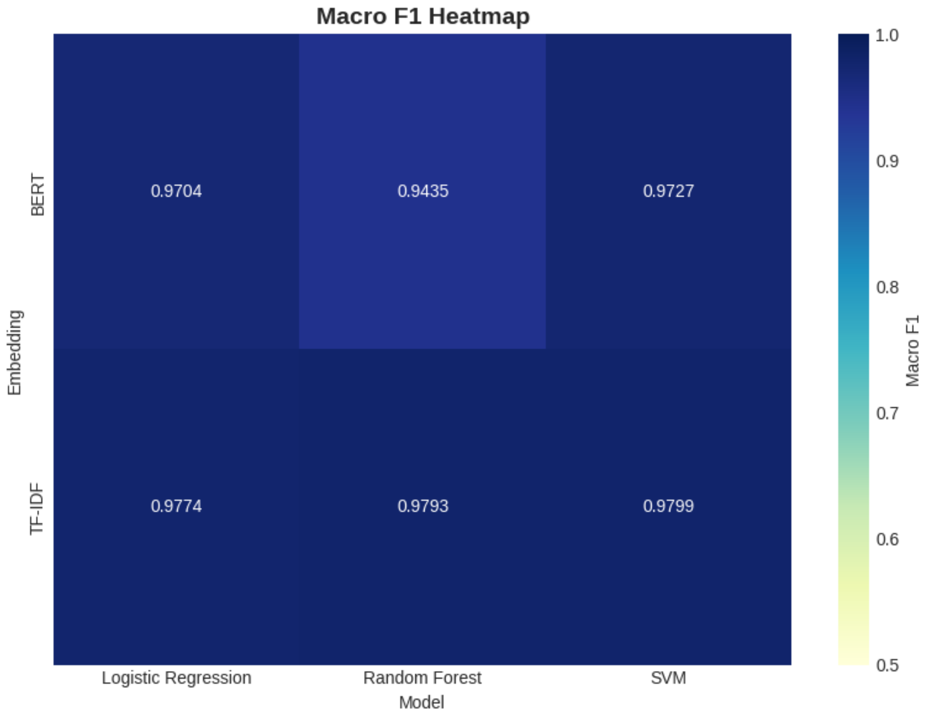
- **Split:** stratified **80/20**
  - Train: **1785**, Test: **447**
  - random_state = 42 (reproducibility)

- **Text representations:**
  - **TF-IDF** (sparse) → shape **(2232, 6985)**
  - **BERT embeddings** (dense) → shape **(2232, 384)**

- **Models tested:**
  - Logistic **Regression**
  - SVM (RBF kernel)
  - Random Forest

Classification Results — Accuracy & Macro-F1





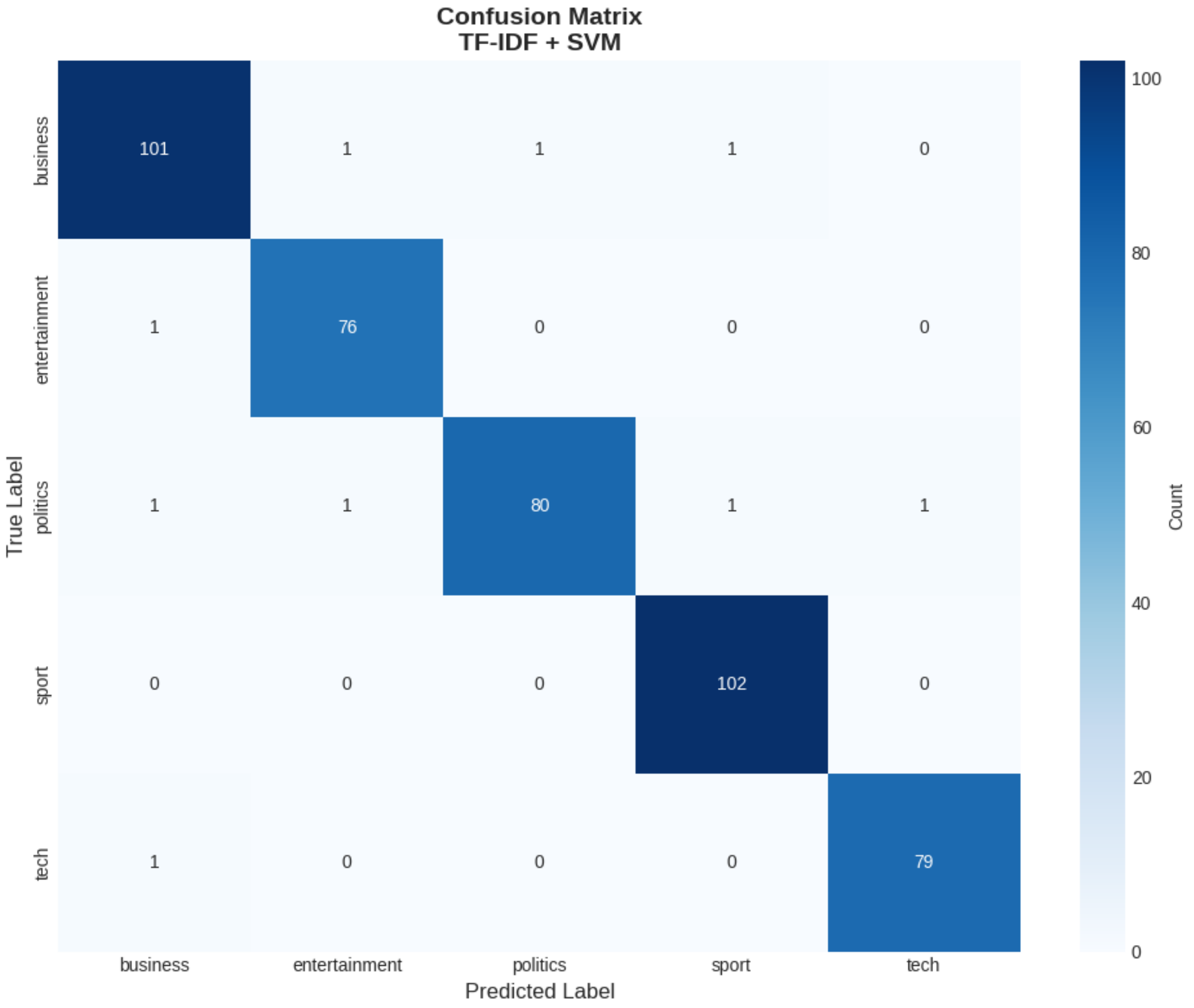
❖ **Best overall: TF-IDF + SVM**
- Accuracy **0.9799**
- Macro-F1 **0.9799**

❖ **Trend:** TF-IDF outperforms BERT across all tested models

❖ **Worst config: BERT + Random Forest (≈ 0.944)**

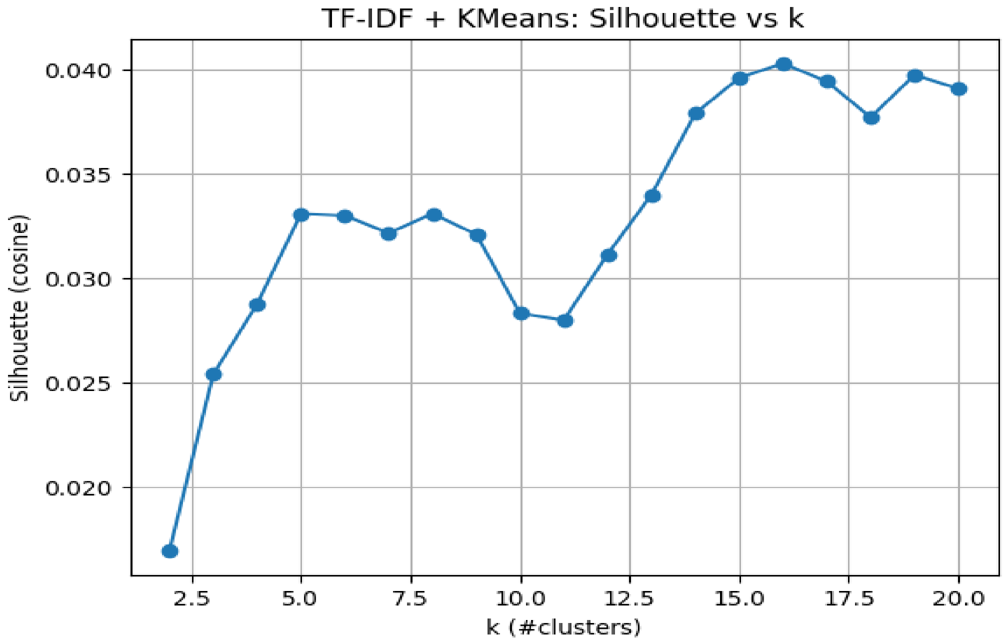
Best Model — Confusion Matrix & Insights

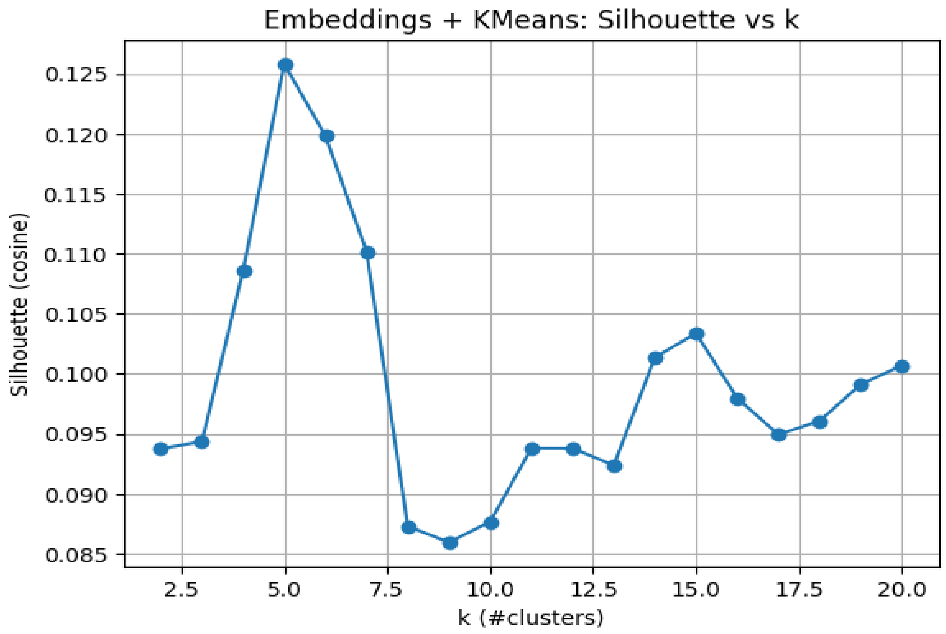
- High performance across all classes (near-diagonal matrix)
- Sport is the easiest class (very high recall)
- Most errors between Business ↔ Politics (shared vocabulary)
- Confusions are limited and consistent with topical overlap



Text Clustering — Setup & Selection of *k*

We evaluate clustering on TF-IDF vs Sentence-Transformer embeddings and select *k* via silhouette analysis.





❖ **Algorithms**

- **KMeans** (main)
- **Agglomerative clustering** (baseline comparison)

❖ **Model selection**

- Silhouette sweep with **k = 2 … 20**
- **Best k (TF-IDF + KMeans): 16**
- **Best k (Embeddings + KMeans): 5** (matches BBC 5 categories)

Clustering Results — Metrics Comparison

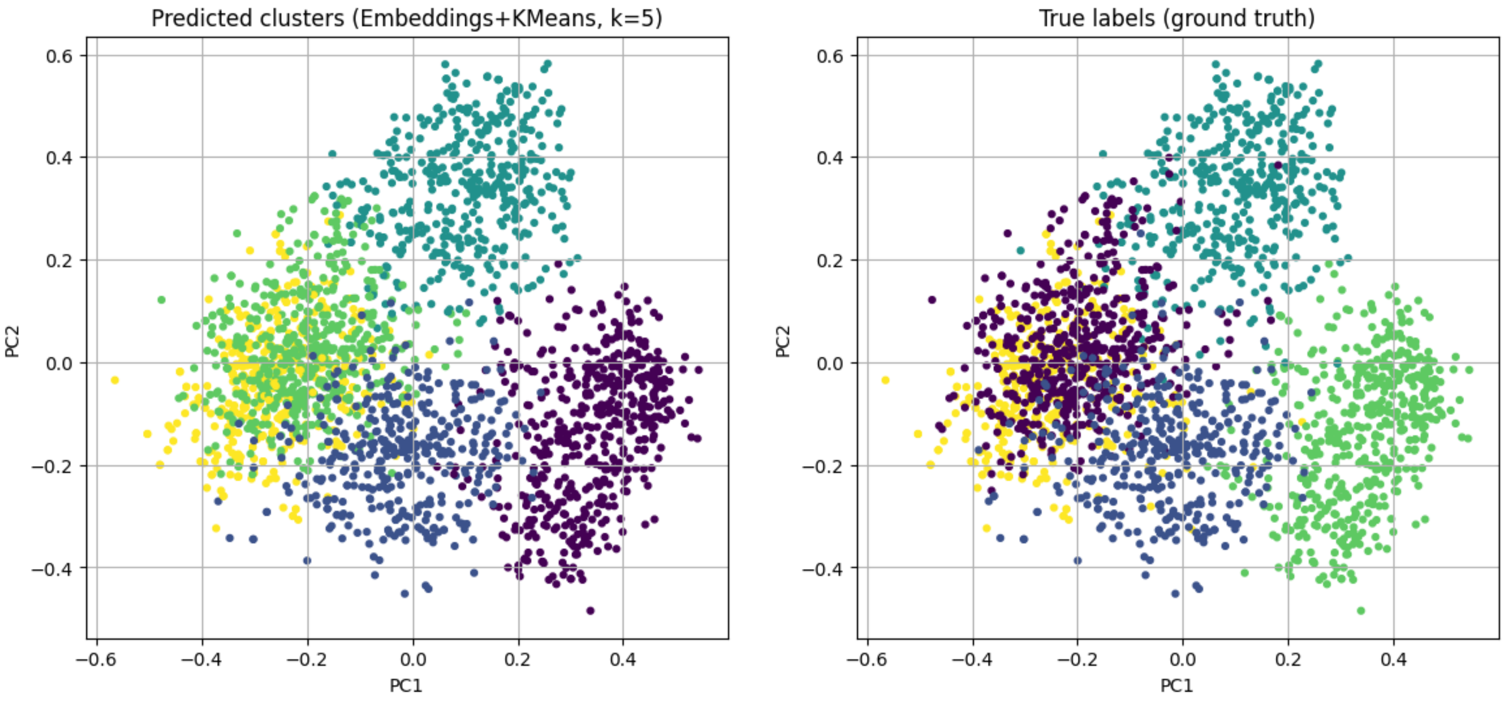
- We report **internal + external metrics**:
  - Silhouette (internal)
  - **ARI / NMI / V - measure** (vs true labels)
  - **Purity + Mapped Macro-F1**

- **Best overall: Embeddings + KMeans (k=5)**
  - Silhouette **0.1258**
  - **ARI 0.9035, NMI 0.8744**
  - Purity **0.9592**, Macro-F1(mapped) **0.9582**

- TF-IDF tends to prefer **higher k (16)** → finer lexical clusters

	model	k	silhouette	ARI	NMI	V_measure	purity	F1_macro_mapped
2	Embeddings + KMeans	5	0.125800	0.903473	0.874429	0.874429	0.959229	0.958220
3	Embeddings + Agglomerative(cosine,avg)	5	0.079821	0.499457	0.603504	0.603504	0.604839	0.476297
1	TF-IDF(SVD200) + Agglomerative(ward)	16	0.058099	0.374230	0.566989	0.566989	0.851254	0.846131
0	TF-IDF + KMeans	16	0.040268	0.275622	0.535511	0.535511	0.793907	0.800028

Best Clustering — PCA Visualization & Cluster Purity



Cluster (k=5)	0	1	2	3	4
True label					
business	0.39	0.19	2.51	93.42	3.48
entertainment	0.00	94.82	0.78	0.78	3.63
politics	0.72	0.24	95.20	2.88	0.96
sport	99.22	0.59	0.20	0.00	0.00
tech	0.75	2.00	0.00	0.50	96.76

- ❖ PCA shows **clear separation** between clusters in embedding space
- ❖ **Strong alignment** with ground truth labels (unsupervised recovery of classes)
- ❖ Crosstab highlights **high cluster homogeneity**
- ❖ Cluster sizes are **balanced** (no dominant/degenerate clusters)

Conclusions & Key Takeaways

Text representation plays a crucial role and its effectiveness strongly depends on the task.

- ❖ **Text representation has a significant impact** on both classification and clustering performance.

- ❖ **Supervised classification:**
  - **TF-IDF + SVM** achieved the best results
  - Lexical features are highly discriminative for news categories.

- ❖ **Unsupervised clustering:**
  - **Sentence embeddings + KMeans** produced the most coherent clusters
  - Dense representations better capture **high-level semantic similarity**.

- ❖ **No single representation is universally optimal:**
  - Sparse models excel in supervised settings
  - Dense embeddings are more effective for semantic structure discovery.

- ❖ **Representation choice should be task-driven**, balancing interpretability and semantic richness

THANK YOU !!